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Human Designers' Dynamic Confidence and Decision-Making When Working With More Than One Artificial Intelligence

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As artificial intelligence (AI) systems become increasingly capable of performing design tasks, they are expected to be deployed to assist human designers' decision-making in a greater variety of ways. For complex design problems such as those with multiple objectives, one AI may not always perform its expected accuracy due to the complexity of decision-making, and therefore, multiple AIs may be implemented to provide design suggestions. For such assistance to be productive, human designers must develop appropriate confidence in each AI and in themselves and accept or reject AI inputs accordingly. This work conducts a human subjects experiment to examine the development of a human designer's confidence in each AI and self-confidence throughout decision-making assisted by two AIs and how these confidences influence the decision to accept AI inputs. Major findings demonstrate severe decreases in a human designer's confidence especially when working with one or more low-performing AI teammates and/or receiving negative feedback. Additionally, a human designer's decision to accept AI suggestions depends on their self-confidence and confidence in one of the two AIs. Finally, an additional AI does not increase a human designer's likelihood of conforming to AI suggestions. Therefore, in comparison to a scenario with one AI, the results in this work caution against the implementation of an additional AI to AI-assisted decision-making scenarios. The insights also inform the design and management of human–AI teams to improve the outcome of AI-assisted decision-making. [DOI: 10.1115/1.4064565]

Keywords: artificial intelligence, collaborative design, design methodology, design process, design teams

1 Introduction

As artificial intelligence (AI) systems continue to prove their usefulness in engineering design [1–8], there are increasing efforts to leverage their strengths to enhance human designers' performance [9–14]. Based on insights pulled from big data, AIs often provide design assistance via suggestion(s). Existing data-driven tools and AI agents in design offer suggestions in various steps of the design process including but not limited to process-related advice [12], analogies that may spark new design ideas [15], novel ideas yet to be reached by human designers [7], and next design action to take [8,14].

In the current state of human–AI design collaboration, AI-assisted decision-making is a common scenario where human designers have the authority to either accept or reject the AI suggestions and make the final design decision. Given this, human–AI design teams are not able to reach their full potential if human designers do not successfully choose when to accept and reject AI suggestions. When inappropriately accepted or rejected, resulting design solutions may even be worse than those from only human designers or AI. Therefore, much literature on human–AI teaming both in and outside of design has paid great attention to human trust in AI as a crucial factor that contributes to the decision to rely/not rely on AI input [16–25]. In design, Chong et al. investigated the changes in a human designer's confidence in their AI teammate and self-confidence (antecedents of trust chosen among the various confounding factors of human trust [26,27]) in an AI-assisted decision-making scenario with a truss design task [25]. They then found how these confidences influence a human designer's reliance on the AI teammate.

Despite this research's attention on the dynamics between a human designer and an AI during AI-assisted decision-making, there has not been much work, particularly in design, on human–AI teams with more than two members. However, the increase in the accessibility and performance of AI will surely give rise to larger and more diverse human–AI design teams to solve complex engineering design problems. Often, complex engineering design problems (e.g., ill-structured problems, multi-objective optimization problems) are decomposed into multiple subproblems to be tackled effectively [28–32]. Additionally, in difficult situations,

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there may be fast and possibly conflicting information given to a designer (or engineer). These scenarios highlight the need for well-composed human–AI teams with more members to solve design problems in an innovative and timely manner. For example, an AI may be implemented for each subproblem, making up multiple AIs providing suggestions to the human designer(s). Therefore, for human designers and AIs to successfully solve complex design problems together, it is crucial to investigate human designers' confidence dynamics and AI-assisted decision-making patterns in larger teams along with one-human/one-AI design teams.

This work is one formal study to investigate AI-assisted decision-making scenarios with more than one AI by considering one-human/two-AI teams working on a configuration design task. Specifically, it examines the changes in a human designer's confidence in the two-AI teammates (i.e., AI #1 and AI #2) and self-confidence, as well as the impact of this confidence on the decision to accept or reject AI suggestions. There are three conditions with different combinations of the two-AI teammates' performance: AI #1 and AI #2 are (1) both high-performing, (2) both low-performing, or (3) one high-performing and one low-performing. These conditions account for the imperfections in real-world AI systems that depend on various factors such as the training data, the situation, and the algorithm itself. Particularly in multi-objective problems, AIs are likely to provide conflicting suggestions that may be "good" according to one AI's objective but not according to other AIs' objectives. The design task (i.e., truss design) and the procedure in this study resemble those from the one-human/one-AI team study by Chong et al. [25] to enable a comparison of the results. Given the prior works that showed the effects of team composition and size on human–AI team performance, member's team cognition, situation awareness, and trust [33,34], results with one-human/two-AI design teams are expected to be significantly different from those with one-human/one-AI teams. This work focuses on the following research questions (RQ):

(1) How does a human designer's confidence in the two-AI teammates and self-confidence change with different AI performances?

Chong et al. demonstrated how a human designer's confidence in an AI and self-confidence change with the AI's performance [25]. Specifically, poor AI performance was shown to have a detrimental impact on a human designer's confidence in that AI. A human designer's self-confidence generally presented a decreasing trend regardless of AI performance. Similarly in this work, it is expected for a human designer's confidence in the two-AI teammates to decrease with poor AI performance: the confidence in each AI may decrease when that AI performs poorly. With the additional AI in this study, the rate of this decrease may differ depending on the two-AI teammates' combined performance. This hypothesis is supported by Bandura's work which identified that a team's prior performance is a significant factor in team confidence and/or collective efficacy [35]. Furthermore, a human designer's self-confidence, like in Ref. [25], is hypothesized to show a decreasing trend regardless of AI performance.

(2) How do positive and negative feedback on a human designer's decision affect their confidence in each AI and their self-confidence?

The results in Ref. [25] showed how feedback on a human designer's chosen design action corresponds to their confidence in the provider of the chosen action, as well as their self-confidence. A similar result is expected in this work.

(3) How are these confidences each correlated to a human designer's probability of accepting AI #1's suggestions and the probability of accepting AI #2's suggestions?

Both confidence in AI and self-confidence were found to influence a human designer's decision to accept or reject an AI suggestion [25]. In this work, all confidences are again expected to influence a human designer's decisions. Especially, confidence in AI is expected to have a more significant correlation than self-confidence because the truss design task in this study is an intellective task with correct answers [36]. In addition, designers are hypothesized to make rational decisions: the probability of accepting an AI's suggestion is affected by their self-confidence and confidence in that AI, not the other AI. For example, the probability of accepting AI #1's suggestion is hypothesized to be affected by self-confidence and confidence in AI #1 (and not in confidence in AI #2).

(4) Is a human designer more likely to accept a design suggestion that two AIs agree on than one provided by a single AI?

Prior psychology literature has shown that people in group settings tend to conform to the norms of the majority [37,38]. This finding leads to a hypothesis that a human designer will more likely conform to a design suggestion that two AIs agree on (i.e., AI #1 and AI #2 provide the same suggestion) than one provided by one AI. Furthermore, based on prior work that found that humans tend to conform to recommendations more with information overload [39], a human designer is expected to be more likely to conform to any AI suggestions when receiving multiple suggestions from two-AI teammates compared to when receiving a suggestion from one-AI teammate.

2 Methods

2.1 Human Subjects Study. A human subjects study is designed and conducted to investigate how a human designer makes decisions with AI assistance in a one-human/two-AI design team. Specifically, the study focuses on the changes in a human designer's confidence in each of their two-AI teammates and self-confidence, as well as their acceptance of AI suggestions over the course of 30 truss design problems. There are three experimental conditions that vary by the performance of the two-AI teammates, which are included to study the effects of AI performance or the combination of AI performance on a human designer's confidence. AI #1 and AI #2 are given gender-neutral names in the study: Taylor and Alex, respectively, to emphasize their presence as an actual teammate rather than a mere tool while reducing gender bias; however, participants are told that they are AI agents. In this paper, they are referred to as AI #1 and AI #2.

2.1.1 Participants. About 136 participants were recruited for (and completed) the experiment in accordance with a protocol approved by Carnegie Mellon University's Institutional Review Board. All participants are first-year undergraduate students studying Mechanical Engineering and enrolled in the Fundamentals of Mechanical Engineering course at Carnegie Mellon University. Just prior to the experiment, the course covered mechanics, specifically truss systems, therefore ensuring that all participants have a base-level understanding of truss design to successfully complete the experimental task (refer to Sec. 2.1.2 for more information about the task). Informed consent is obtained from all participants before they participate in the experiment. 6 of the 136 participants' data are filtered out during the data cleaning process; therefore, data from 130 participants are included in the analyses in this work.

2.1.2 Experimental Task. In the study, the participants are asked to work on 33 truss design problems (three for practice and the rest for data collection) with AI assistance, a design task that resembles the one from [25]. In each problem, the participants are given a unique truss state (not sequential design) and are asked to make the best next design action that would be advantageous in maximizing the strength-to-weight ratio (SWR) of the truss (see Fig. 1 for an example problem). Possible design actions are limited to "Delete Node," "Add Member," "Delete Member," "Decrease Member Thickness," and "Increase Member Thickness," the latter two at defined increments. This is to limit the size of the design space to allow the evaluation of all the possible actions per problem. This design task is suitable for this experiment

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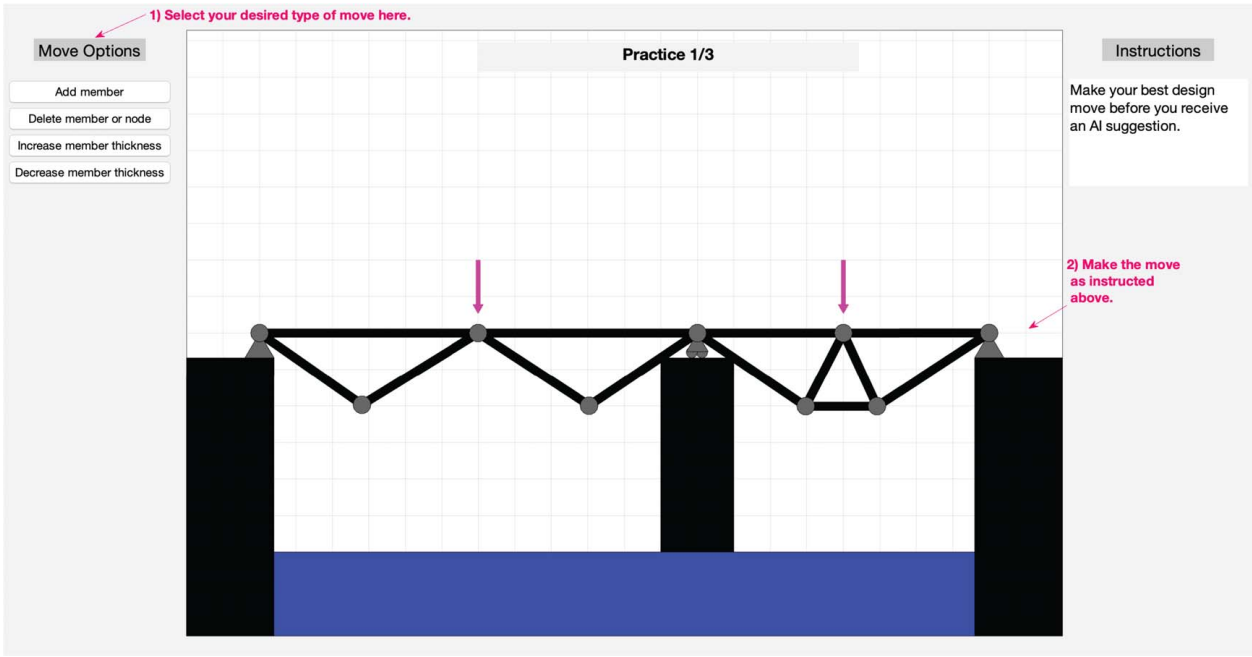


Fig. 1 Example problem in the truss design task

because it is well-defined, as well as solvable independently by a human designer while complex enough to benefit from AI suggestions.

Data collection over 33 truss design problems captures a human designer's changing confidence in and use of AI teammates' suggestions in a continuous, rather than one-time, AI-assisted decision-making scenario. The participants in this experiment all receive the same 33 truss design problems in the same order while the order of good and bad AI suggestions is randomized. For example, in a Hi–Hi condition, while each AI provides a good suggestion 80% of the time, when these good suggestions will be given is randomized for each participant. This randomization is implemented to avoid the ordering effect of the results. A major difference in this work from [25] is the additional AI. While each participant collaborated with a single AI teammate that provided a suggestion per problem in the study by Ref. [25], each participant in this work collaborated with two-AI teammates who each provided a suggestion (total of two suggestions) per problem.

The algorithm behind the AI teammates in this study is a depth-3 tree search method developed by Chong et al. [25] by utilizing the large, sequential truss design dataset collected by Raina, et al. [8]. Given a truss design state, the algorithm first identifies all the possible actions from that state. Then, it searches three steps ahead from each of those actions, selecting the top five actions in each step. The top five actions are selected by the likelihood of a human design making the action based on Raina, et al.'s sequential truss design dataset [8]. The goodness score of each initial action is calculated by taking the weighted sum of the SWR of the resulting states at depth 3. In this work, "good" or advantageous actions are defined as those that have goodness scores that are greater than one-half of the maximum goodness score (belonging to the best action) in that problem, and the rest are considered "bad" or disadvantageous actions. During the truss design problem selection process for this experiment, it is ensured that at least two advantageous and two disadvantageous actions can be made in each problem. This is not only to ensure consistency in problem difficulty and variety of possible actions to choose from but also to control for the experimental conditions described in the next section. In addition to being used to set up the AI teammates, this algorithm is used to evaluate and provide feedback on any design actions that are made or selected in this experiment. For further details about the algorithm, refer to [25].

2.1.3 Experimental Conditions. 136 participants are randomly assigned to one of the three condition groups (45, 45, and 46 participants, respectively) determined by the performance of the two-AI teammates: AI #1 and AI #2. AI #1's suggestion is always shown on the left of AI #2's suggestion. The first condition is the Hi–Hi condition, where the two-AI teammates are high-performing. In this work, high-performing AI means that it gives a "good" suggestion 80% of the time (24 out of the 30 truss design problems) and "bad" suggestions for the remaining 20%. These two percentages (80% and 20%) are selected based on the number of problems (i.e., 30) and what mimics the imperfections of AI systems in the real-world. No AI is perfect (100%), and 90% will only give three "bad" suggestions out of 30 which is not enough for the participants to easily recognize; therefore, 80% is selected. In the same way, 20% is chosen to make the low performance of the AI obvious but not too obvious. "Good" suggestions are selected from the top two advantageous actions among all the possible actions with goodness scores higher than 1/2 of the maximum score in the problem. Similarly, "bad" suggestions are selected from the bottom two disadvantageous actions among all the possible actions with goodness scores lower than 1/2 of the maximum score. The second condition is the Lo–Lo condition, where the two-AI teammates are low-performing, providing a "good" suggestion 20% of the time (6 out of the 30 truss design problems). Finally, the third condition was the Hi–Lo condition, where AI #1 is high-performing while AI #2 is low-performing. The reverse scenario where AI #1 is low-performing and AI #2 is high-performing is not included as a condition because the visual order is not expected to have a significant impact on the results.

In all three conditions, the AI suggestion agreement (i.e., when the two-AI teammates suggest the same design action) is controlled to happen 40% of the time; the two AIs provide identical suggestions in 12 of the 30 problems. This controlled agreement rate is possible because there is more than one advantageous and more than one disadvantageous possible action per problem. For example, in the Hi–Hi condition, the availability of multiple advantageous and disadvantageous actions enables the two-AI teammates to suggest different yet advantageous actions. In this way, the two-AI suggestions can disagree 60% of the time while maintaining the AIs' high performance. Additionally, the experiment controlled for the task (i.e., all the problems and their order), procedure, and the participants' level of expertise.

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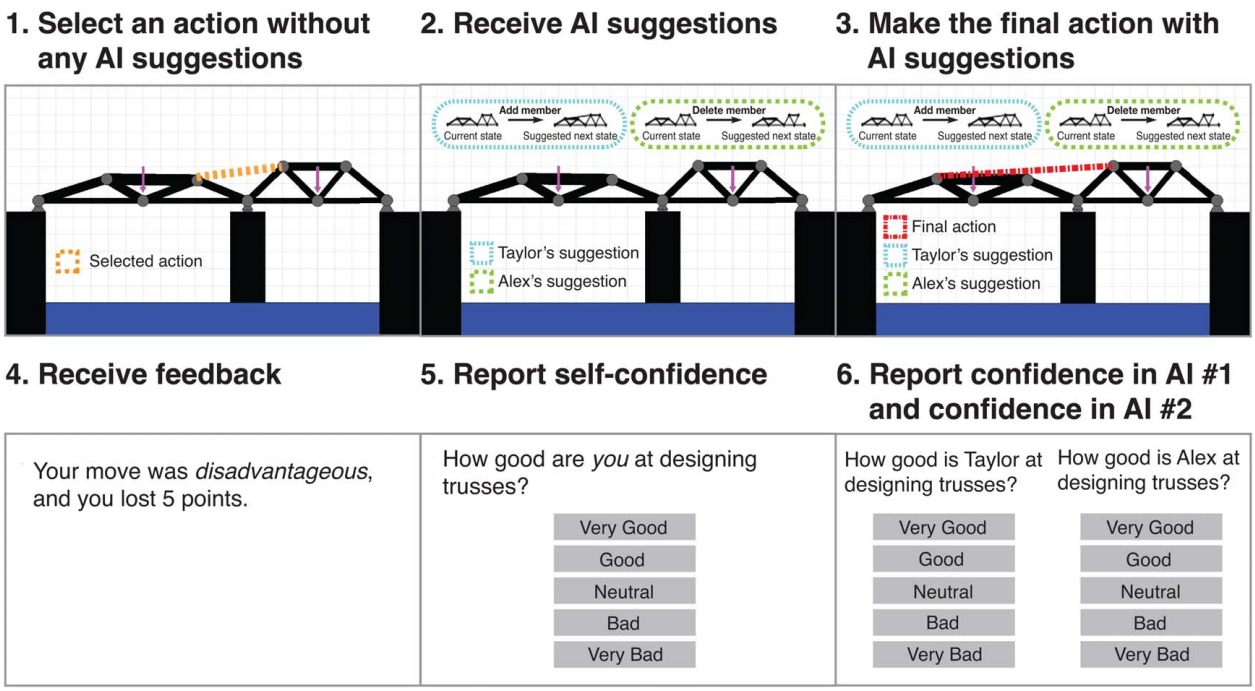


Fig. 2 Experimental procedure

2.1.4 Procedure. The experiment is conducted in a computer lab during the four separate recitation sessions of the Fundamentals of Mechanical Engineering course. The four sessions are held consecutively on the same day. There are 36, 36, 35, and 29 participants in the sessions in order, and in each session, the participants are randomly assigned to one of the three condition groups, adding up to 46, 45, and 45 participants in Hi–Hi, Lo–Lo, and Hi–Lo conditions, respectively. Each participant receives a $10 compensation for their participation in the study.

All the participants are assigned to a computer on which the entire process of these two percentage (80% and 20%) experiment is conducted. The necessary experimental interface and links are all predownloaded on the computers. The participants first receive a brief introduction of the experiment as a human–AI team design study without any details and are asked to fill out an online informed consent. Then, they are asked to carefully read through the instructions on the experimental task and the use of the interface. They are not told of the experimental condition they are in or of whether the two-AI teammates are high- or low-performing. A point system is introduced to the participants to incentivize them throughout the experiment: those who achieve a score above 80 will receive an additional $5 monetary prize. This threshold is determined via a pilot study as an excellent score that is achievable but infrequently achieved. Once the above steps are completed, the participants are instructed to begin the task.

For all the 33 truss design problems, the participants follow the same procedure (see Fig. 2). This procedure is identical to that in the study by Ref. [25] except for the changes with the additional AI teammate. First, the participants select their best next action for the given truss state without any AI suggestions. This selected action is called the unassisted action in this work because it is an independent action made without any assistance. This design of the procedure not only prevents the participants from entirely depending on the AI suggestions but also allows data collection of the participants' proficiency without the AIs. Then, the participants receive suggestions from the two-AI teammates at the same time (AI #1's suggestion on the left and AI #2's suggestion on the right) and make a final design action (i.e., assisted action) considering the suggestions. This final action can be one of the AI-suggested actions or any action different from those. Once the final action is made, feedback about the "goodness" of the final action (either positive or negative) is given to the participants. Positive feedback indicates that their final action is advantageous and that they are gaining 5 points, while negative feedback indicates that it is disadvantageous and that they are losing 5 points. The feedback is determined by the depth-3 tree search algorithm discussed earlier in the *Experimental Task* section. Finally, the participants report their self-confidence, confidence in AI #1, and confidence in AI #2 by answering the following questions in order: How good are/is you/Taylor (AI #1)/Alex (AI #2) at designing trusses? For each question, the answer is reported in a 5-point Likert scale: very good, good, neutral, bad, and very bad, which are, respectively, quantified as 1, 0.75, 0.5, 0.25, and 0.

2.2 Confidence Model. The quantitative model of human confidence developed by Chong et al. [25] is extended and used in this work to answer RQ2: How do positive and negative feedback on a human designer's decision affect their confidence in each AI and self-confidence? Fitted to the human data from the experiment, this model captures the relationship between eight different experiences during the AI-assisted decision-making scenario with two AIs (i.e., different combinations of human decision and corresponding feedback) on the participants' confidence in AI #1, confidence in AI #2, and self-confidence.

2.2.1 Model Description. The confidence model calculates human confidence at a given trial based on experience, accumulated confidence, and bias [23,24]. Chong et al. [24] to find explanations for how the model was developed. The model is applied to any human confidence regardless of the subject of confidence; therefore, the model is used for all three confidences that are being studied in this work: confidence in AI #1, confidence in AI #2, and self-confidence. The general model equation [23]:

$$C(n+1)=C(n)+\alpha_e[E(n)-C(n)]+\alpha_a[A(n)-C(n)]+\alpha_b[B(n)-C(n)] \quad (1)$$

where $C(n), E(n), A(n), B(n), \alpha_e, \alpha_a, \alpha_b \in [0,1]$, contains the three factors of human confidence mentioned above: experience ($E(n)$), accumulated confidence ($A(n)$), and bias ($B(n)$). The weighted sum of the differences between each of these factors and confidence at trial n ($C(n)$) represents the change in confidence from trial n to

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$n+1$. The subtraction of the three factors from confidence at trial n can be computed given the assumption of this model that the values of the experience, accumulated confidence, and bias terms represent the appropriate confidence generated from those factors. Therefore, the confidence value at trial $n+1$ ($C(n+1)$) can be deduced. The weights α_e, α_a, and α_a in the model are the rate factors of experience, accumulated confidence, and bias, respectively.

The experience term ($E(n)$) in Eq. (1) denotes the experience that humans have at trial n. In Ref. [25], there were four different possible experiences given its design collaboration scenario with a single AI teammate. In this study, with an additional AI teammate, there are eight different experiences, which are:

(1) Accept AI #1's suggestion, reject AI #2's suggestion, then receive positive feedback;
(2) Accept AI #1's suggestion, reject AI #2's suggestion, then receive negative feedback;
(3) Accept AI #2's suggestion, reject AI #1's suggestion, then receive positive feedback;
(4) Accept AI #2's suggestion, reject AI #1's suggestion, then receive negative feedback;
(5) Accept both AIs' suggestions, then receive positive feedback;
(6) Accept both AIs' suggestions, then receive negative feedback;
(7) Reject both AIs' suggestions, then receive positive feedback;
(8) Reject both AIs' suggestions, then receive negative feedback.

The experience term is computed by summing the eight sub-experience terms (corresponding to the above list of experiences) with their respective weights (ω_1, ω_2, ω_3, ω_4, ω_5, ω_6, ω_7, and ω_8):

$$E(n) = \sum_{i=1}^{8} \omega_i e_i(n) \tag{2}$$

where $e_i(n) = 0$ or 1, $\omega_i \in [0,\ 1]$, and $\sum_{n=1}^{8} e_n(n) = 1$. Fitted parameter values of the eight weights show the impact of their corresponding experiences on confidence.

The accumulated confidence term at trial n ($A(n)$) represents the time-discounted, weighted sum of the confidences before trial n. The bias term at trial n ($B(n)$) accounts for the general bias participants have towards AI systems, not the specific AI that they are working with. Therefore, the value of this term is set to the confidence reported after the three practice problems and right before the first problem and is kept constant throughout the experiment. For further descriptions of the accumulated confidence $A(n)$ and bias $B(n)$ terms, refer to [25].

2.2.2 Parameter Fitting. Trust region reflective algorithm [40] is used for parameter estimation. This algorithm, when provided with initial values, iteratively refines the parameters and computes the error until the optimal values are found. The parameter estimation is repeated with different initial values to confirm that the optimal parameter results do not fluctuate. Table 1 shows the optimal parameter values fitted to the human data from the experiment. The overall optimal parameter values are found for each of the three confidences: human confidence in AI #1, confidence in AI #2, and self-confidence; α_e, α_a, α_b are the rate factors of experience, accumulated confidence, and bias terms, respectively; ω_1, ω_2, ω_3, ω_4, ω_5, ω_6, ω_7, and ω_8 are the experience impact factors of the eight different experiences introduced in the previous section. γ is the time discounting factor of the accumulated confidence term. In the Results section, the optimal parameter values for each confidence are found separately in the three experimental conditions to explore how the impact of different experiences on the confidence varies according to the performance of the AI teammates.

For every row of the parameter estimation results in Table 1, a robustness test is conducted by estimating the parameters using 80% of the corresponding experimental data. In each robustness test, the parameter estimation is repeated 100 times, every time with a new randomly selected 80% of the data. The initial values are set to the values in Table 1. All the robustness tests show consistent results among the 100 trials with minor differences (below 0.18 mean absolute deviation from the fitting results in Table 1), validating that the parameter estimation results are robust.

Having found the optimal parameter values along with the initial confidence data from the experiment, the model can iteratively compute human confidence in AI #1, confidence in AI #2, and self-confidence in each trial of the experiment. Table 1 displays the mean squared error (MSE) and adjusted R-squared results of the model fitting for each confidence.

3 Results

Eliminating the six participants who failed to adhere to the instructions and/or to complete the experiment, there are 42, 45, and 43 participants in the Hi–Hi, Lo–Lo, and Hi–Lo conditions, respectively. The results from the data of these 130 participants are described in this section.

3.1 Change in Human Confidence With Different Artificial Intelligence Performance Combinations. This section presents how human confidence in AI #1, confidence in AI #2, and self-confidence change over the course of the AI-assisted decision-making design task with two-AI teammates depending on the AIs' performance combination, which is determined by the experimental conditions. Figure 3 displays the results. The data points in the plots in Fig. 3 are the trial-by-trial, average confidence values of the participants in that condition. The slopes of linear regression show the general changes in confidence over the 30 problems.

Figures 3(*a*) and 3(*b*) show that both the participants' confidence in AI #1 and confidence in AI #2 decrease throughout collaboration in all three conditions (F-test; $p < 0.01$ for all six slopes). This result is interesting because it means that regardless of the two-AI teammates' performance, the participants lose confidence in both AIs over time. In addition, the participants' average confidence in each AI teammate during the experiment differs between the conditions. In the Hi–Hi condition, the average confidence in AI #1 and that in AI #2 are both 0.46. In the Lo–Lo condition, they are 0.33 and 0.32, respectively. Finally, in the Hi–Lo condition, they are 0.44 and 0.36. In any of these conditions, the difference in the average confidence between a high-performing AI and a low-performing AI is always statistically significant (t-test; $p < 0.01$), while the difference in the average confidence between two high-

Table 1 Parameter estimation results for the confidence model

Model	Optimal parameter values												MSE	R^2_{adj}
	α_e	α_a	α_b	ω_1	ω_2	ω_3	ω_4	ω_5	ω_6	ω_7	ω_8	γ		
Confidence in AI #1	0.44	0.39	~0	0.96	~0	0.31	0.34	~1	~0	0.28	0.32	0.33	3.1e-4	0.89
Confidence in AI #2	0.49	0.26	~0	0.44	0.24	0.84	0.14	~1	~0	0.29	0.32	0.33	9.6e-4	0.61
Self-confidence	0.25	0.41	~0	0.55	~0	0.75	~0	0.76	~0	0.78	~0	0.16	7.0e-4	0.69

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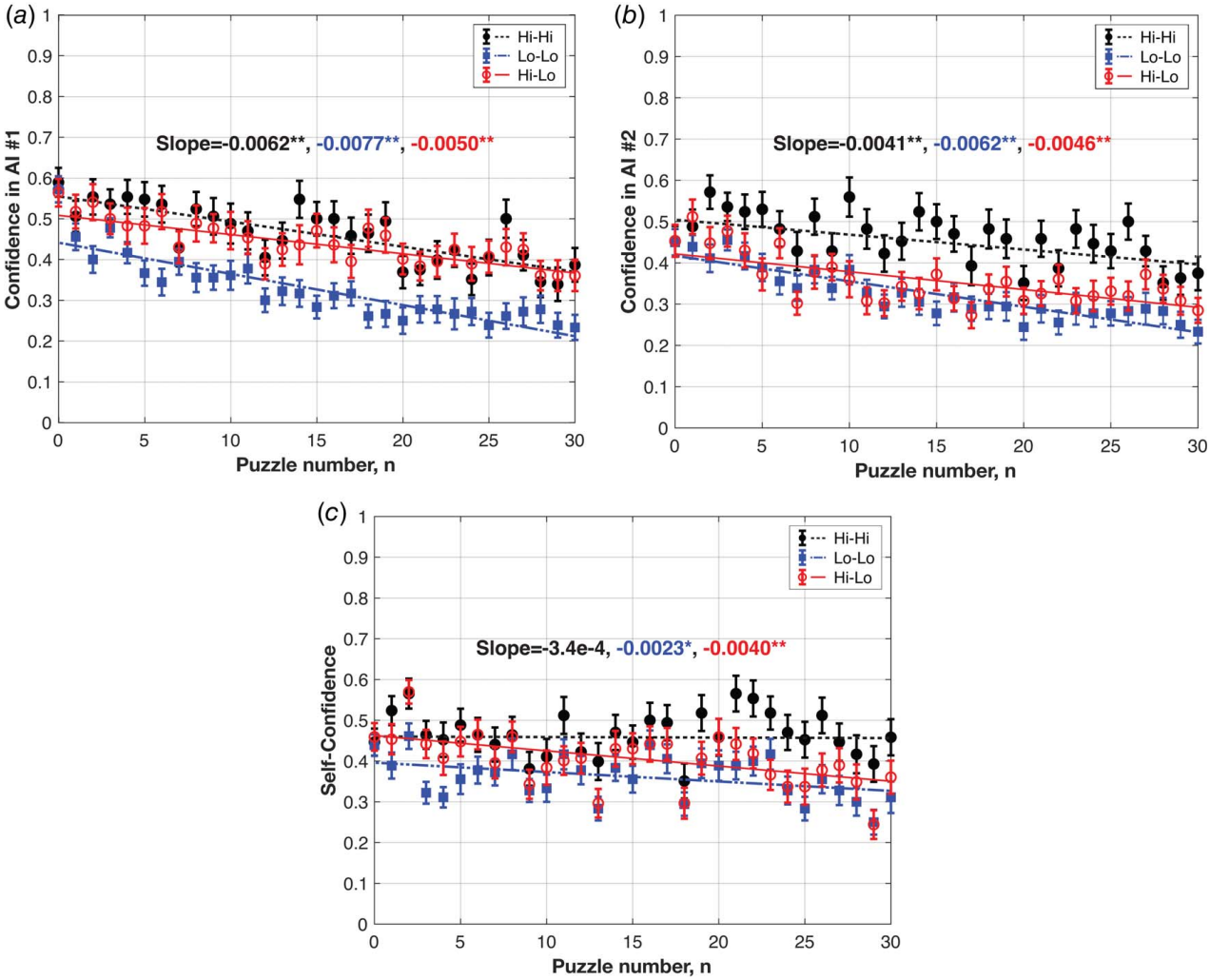


Fig. 3 Changes in the participants' (*a*) confidence in AI #1, (*b*) 387 confidence in AI #2, and (c) self-confidence. * indicates significance at the 5% 388 level, and ** indicates significance at the 1% level

performing AIs or two low-performing AIs is always statistically insignificant (*t*-test; $p > 0.05$). These results illustrate that the participants have a higher average confidence in a high-performing AI teammate and vice versa, especially when the two AIs are both high-performing or low-performing. This result is consistent when one AI is high-performing and another is low-performing, although the participants may not be as confident in the high-performing AI as when both AIs are high-performing and not as unconfident in the low-performing AI as when both AIs are low-performing.

Figure 3(*c*) shows the participants' self-confidence throughout the experiment. The participants' self-confidence does not significantly change over time in the Hi–Hi condition ($p = 0.76$), while it decreases in the Lo–Lo and Hi–Lo conditions ($p < 0.05$ and <0.01, respectively). These results mean that when both AIs are performing well, human designers' confidence in their own abilities remains stable. However, when one or both AIs perform poorly, human self-confidence drops. The participants also demonstrate varying levels of average self-confidence among the three conditions. In the Hi–Hi, Lo–Lo, and Hi–Lo conditions, their average self-confidence during the experiment is 0.47, 0.36, and 0.40, respectively. It is notable that the greater the number of low-performing AIs, the lower the participants' average self-confidence.

3.2 Impact of Feedback on Human Decision on Human Confidence. There are eight different types of experiences a human designer can undergo as they are working with two-AI teammates. These experiences entail different combinations of human decision (i.e., whether they accept their own design action or one of the AIs' suggestions) and corresponding feedback (positive or negative). Refer to the Methods section under *Model Description* for the list of these experiences. To find the impact of these experiences on the participants' confidence, the three confidence models (confidence in AI #1, confidence in AI #2, and self-confidence) are each fitted to the human data from the experiment. The resulting parameter values of $\omega_{1\sim8}$ reveal the impact of the corresponding types of experiences on the confidences. Table 2 illustrates these results in the three experimental conditions. The arrows represent whether the corresponding confidence increases or decreases because of each type of experience.

After receiving any negative feedback (e_2, e_4, e_6, and e_8), the participants' confidence in AI #1, confidence in AI #2, and self-confidence all decreased. This means that regardless of whose design action the negative feedback is for (e.g., human designer, AI #1, or AI #2), the participants lose confidence in all members of the team including themselves. This result is consistent among all three experimental conditions.

The impact of positive feedback on the participants' confidence in AI #1 and confidence in AI #2 depends on the subject of the feedback (depending on the human designer's final decision). First, when the participants accept either one of the AIs' suggestions and receive positive feedback (e_1 and e_3), they gain confidence in the accepted AI but lose confidence in the other AI. This result

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Table 2 Relationship between various types of experiences and the participants' confidences

Experience						
Accepted suggestion	Feedback	Notation	Condition	Confidence in AI #1	Confidence in AI #2	Self-confidence
Only AI #1's	+	e_1	Hi–Hi	▲	▼	▲
			Lo–Lo	▲	▼	▲
			Hi–Lo	▲	▼	—
	−	e_2	Hi–Hi	▼	▼	▼
			Lo–Lo	▼	▼	▼
			Hi–Lo	▼	▼	▼
Only AI #2's	+	e_3	Hi–Hi	▼	▲	▲
			Lo–Lo	▼	▲	▲
			Hi–Lo	—	—	—
	−	e_4	Hi–Hi	▼	▼	▼
			Lo–Lo	▼	▼	▼
			Hi–Lo	▼	▼	▼
Both AIs'	+	e_5	Hi–Hi	▲	▲	▲
			Lo–Lo	▲	▲	▲
			Hi–Lo	▲	▲	▲
	−	e_6	Hi–Hi	▼	▼	▼
			Lo–Lo	▼	▼	▼
			Hi–Lo	▼	▼	▼
Neither AIs'	+	e_7	Hi–Hi	▼	▼	▲
			Lo–Lo	▼	▼	▲
			Hi–Lo	▼	▼	▲
	−	e_8	Hi–Hi	▼	▼	▼
			Lo–Lo	▼	▼	▼
			Hi–Lo	▼	▼	▼

mostly holds true among all the experimental conditions except when positive feedback is given to AI #2's suggestion (e_3) in the Hi–Lo condition, where the participants' confidence in the two-AI teammates remains unaffected by the feedback. Secondly, when the AIs' suggestions agree and are accepted, positive feedback (e_5) expectedly increases the participants' confidence in AI #1 as well as their confidence in AI #2. Similarly, when the agreed suggestions are rejected and the participant's own design action is selected, positive feedback (e_7) decreases the participants' confidence in both AIs. These results are consistent throughout the three conditions of the experiment.

Positive feedback increases the participants' self-confidence in most cases. This result, in combination with the earlier result that any negative feedback decreases the participants' self-confidence, suggests that the change in human designers' self-confidence is heavily determined by the success of their final decision (i.e., selection of the final design action). The only exceptional scenarios are when the participants accept one of the AIs' suggestions and receive positive feedback in the Hi–Lo condition. In these scenarios, the participants' self-confidence is unaffected by positive feedback.

3.3 Impact of Human Confidence on Their Acceptance of Artificial Intelligence Teammates' Suggestions. Having explored how the participants' confidence in the two AI teammates as well as their self-confidence evolved over the course of AI-assisted decision-making in design, this section focuses on the impact of this confidence on the acceptance of the two AI teammates' suggestions. Logistic regression is used where there are three predictor variables (designers' confidence in AI #1, confidence in AI #2, and self-confidence), two binary response variables (accept/reject AI #1's suggestion and accept/reject AI #2's suggestion), and n-1 dummy variables where n is the number of participants in the corresponding condition. The regression results are displayed in Table 3.

The results illustrate that the participants' confidence in AI #1 and self-confidence always affect both the probability of accepting AI #1's suggestion and the probability of accepting AI #2's suggestion. Interestingly, in each condition, the impact of confidence in AI #1 and that of self-confidence on each probability are always in opposite directions, meaning that if confidence in AI #1 has a positive impact, then self-confidence has a negative impact, and vice versa. Furthermore, generally, the participants' confidence in AI

Table 3 Correlation between the participants' confidence and their probability of accepting either AI #1 or AI #2's suggestion

		Correlation coefficient		
	Condition	Confidence in AI #1	Confidence in AI #2	Self-confidence
Probability of accepting AI #1's suggestion	Hi–Hi	0.64[a]	0.72[a]	−0.69[b]
	Lo–Lo	−2.1[b]	−0.33	1.7[b]
	Hi–Lo	2.2[a]	−0.095	−0.59[a]
Probability of accepting AI #2's suggestion	Hi–Hi	−0.78[b]	−0.32	0.47[a]
	Lo–Lo	1.44[b]	0.75	−1.1[b]
	Hi–Lo	1.3[b]	0.43	−0.67[a]

[a]indicates significance at the 5% level.
[b]indicates significance at the 1% level.

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Table 4 Probability of accepting AI suggestion when there is one-AI suggestion versus two-AI suggestions that agree or disagree

Number of AIs and suggestion agreement	Probability of accepting AI suggestion
One AI [25]	0.41
Two AIs and they agree	0.42
Two AIs and they disagree	0.41 = 0.21 (AI #1) + 0.20 (AI #2)

#2 neither affects the probability of accepting AI #1's suggestion nor the probability of accepting AI #2's suggestion. Recalling that the participants' confidence in AI #1 has a major impact on the acceptance probabilities, it can be inferred that the participants tend to base their AI acceptance decisions on their perception of AI #1 rather than their perception of AI #2.

3.4 Impact of Artificial Intelligence Teammates' Suggestion Agreement on Human Acceptance of the Suggestion. This work focuses on an AI-assisted decision-making scenario of one-human/two-AI design teams, as opposed to one-human/one-AI teams in the study by Ref. [25]. Two-AI teammates mean that there are two suggestions to consider, and they may either agree or disagree. Recognizing the prevalence of conformity in group settings [37,38], this agreement between the AI suggestions may impact a human designer's decision to accept or reject them. Therefore, this section compares a human designer's probability of accepting an AI suggestion when there is only one-AI suggestion (based on the data collected by Chong et al. [25]), when there is one suggestion that two-AI teammates agree on, and when there are two different AI suggestions. When there is only one-AI suggestion, the probability of accepting the AI suggestion is calculated as the total number of acceptances divided by the total number of observations. When there are two AIs that provide the same suggestion, the probability of accepting the AI suggestion is the total number of acceptances of either one of the AI suggestions divided by the total number of this scenario where two AIs provide the same suggestion. Similarly, when there are two AIs with two different suggestions, the divisor is the total number of observations where the two AIs provide different suggestions.

Tables 4 and 5 present the results. There are no significant differences in the participants' probability of accepting an AI suggestion between any of the three scenarios (i.e., one-AI suggestion, two-AI suggestions that agree, and two-AI suggestions that disagree) (t-test, $p > 0.05$). An additional AI does not increase nor decrease a human designer's probability of accepting an AI suggestion and neither does the agreement among two-AI suggestions.

4 Discussion

4.1 How Does a Human Designer's Confidence in the Two Artificial Intelligence Teammates and Self-Confidence Change With Different Artificial Intelligence I Performance? The results in this work first reveal a human designer's average confidence in each AI teammate and self-confidence during an AI-assisted decision-making design task depending on the different performance combinations of the AIs (Hi–Hi, Lo–Lo, and Hi–Lo conditions). As expected, the study demonstrates that a designer's average confidence in an AI during teamwork differs from an AI's performance. A designer's average confidence in a high-performing AI is higher than that in a low-performing AI, and particularly higher when both AIs are high-performing. Regarding self-confidence, it is found that a designer's average self-confidence corresponds to the performance combination of the AIs. The average self-confidence is highest to lowest in the order of Hi–Hi, Hi–Lo, and Lo–Lo conditions, meaning that the higher performing the AIs are in tandem (or the more competent the team is), the more confident a designer is in themselves. This result agrees with the consistent finding in prior literature that individual members of a high-performing team are more confident in themselves because they are likely to receive more positive feedback [41–43].

Concerning how a designer's confidence in the AIs changes, they lose both confidence in AI #1 and confidence in AI #2 over time regardless of the performance combination of the AI teammates. With one AI, Chong et al. showed that a human designer's confidence in the AI decreases if the AI is low-performing and remains unchanged if the AI is high-performing [25]. Therefore, an additional AI teammate in this study seems to negatively impact how a designer's confidence in the AIs changes over time: a human designer loses the confidence in both AI teammates even when they are both high-performing. This may be because a human designer's confidence in the two-AI teammates are competing variable. The results in Table 2 show that positive feedback on one AI decreases human confidence in the other AI, scenarios that do not occur with only one-AI teammate. This negative effect of positive feedback on one AI on the perception of another AI likely explains why the confidence in the AIs is lost easily and shows an overall decreasing trend.

Unlike the confidence in the AIs, the overall change in a designer's self-confidence varies with the performance combination of the two-AI teammates. When there is one or more low-performing AI(s) on the team, a human designer loses confidence in themselves

Table 5 Summary of the major findings by research question

	Major findings
RQ1: Performance combination of AI teammates and human confidence	• Average human confidence in each AI teammate and self-confidence throughout teamwork differ by the performance of the two-AI teammates • A human designer's confidence in AI #1 and AI #2 both show a decreasing trend over time regardless of the AI teammates' performance • With one or more low-performing AI teammates, a human designer loses self-confidence over time
RQ2: Feedback and human confidence	• Negative feedback lowers all confidences • Positive feedback on an AI increases human confidence in that AI while decreasing confidence in the other AI. • Positive feedback slightly increases self-confidence
RQ3: Human confidence and decision to accept AI suggestion	• A human designer's confidence in AI #1 and self-confidence, but not confidence in AI #2, influence their decision to accept AI suggestions • The influences of confidence in AI #1 and self-confidence differ. The directions of the two influences depend on the performance combination of the AI teammates and are always in opposite directions
RQ4: Additional AI and conformity	• A human designer is not more likely to conform to two-AI teammates than a single AI teammate

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over the course of the collaboration. In the Lo–Lo condition, a human designer is likely to frequently experience negative feedback on their decisions which explains why their self-confidence decreases, supported by the finding by Chowdhury et al. [41]. However, the decrease in a human designer's self-confidence in the Hi–Lo condition is surprising based on Chong, et al.'s finding where self-confidence did not decrease when working with only one-AI teammate who is low-performing [25]. A plausible explanation for this result is that the different performances between the two-AI teammates in the Hi–Lo condition may negatively impact self-confidence. Table 2 shows that in the Hi–Lo condition, positive feedback on either AI does not increase a human designer's self-confidence as it would in the Hi–Hi and Lo–Lo conditions. In other words, when the two-AI teammates have different performances, good AI performance does not increase a human designer's self-confidence while negative AI performance decreases self-confidence, which leads to an overall decrease in self-confidence over time.

The findings in this section reveal that with two-AI teammates, a human designer's confidence in an AI is prone to decrease over time even with high-performing AIs (in contrast to with one-AI teammate). Furthermore, especially if one or more of the AI teammates are low-performing and/or if their performance is very different, a human designer's self-confidence is also prone to decrease over time. These insights suggest that in design practice, an additional AI teammate should be implemented with caution into a human–AI team as it can hurt a human designer's confidence and therefore trust in the AI teammates. It must be ensured that the new AI teammate is high-performing and similar in performance to the other AI teammate. In addition, when an undesirable decrease in confidence is detected in a human designer and leads to low team performance, the team or the manager of the team must heed the AI teammates' performance as a potential cause.

4.2 How Do Positive and Negative Feedback on a Human Designer's Decision Affect Their Confidence in Each Artificial Intelligence and Self-Confidence? Having explored the effect of the performance combination of the two-AI teammates on a designer's confidence in the AIs and self-confidence, this study investigates the impact of the various types of experiences (whose design action is selected and the corresponding feedback) on human confidence. The results first discover the detrimental impact of negative feedback. No matter whether the negative feedback is directed towards—AI #1, AI #2, or the human designer—the designer's confidence in AI #1, confidence in AI #2, and self-confidence all decrease, as shown in Table 2. This result is consistent with a one-AI teammate scenario in Ref. [25] in which negative feedback decreased a designer's confidence in the AI teammate, as well as their self-confidence. Such a significant effect of negative feedback on a human designer's confidence may be attributed to loss aversion, a human cognitive bias in which the pain of a negative experience is much stronger than the pleasure of a positive experience [44].

Positive feedback has a milder and perhaps more logical impact on a human designer's confidence. When a designer accepts either one of the AIs' suggestions and receives positive feedback, they become more confident in that AI but less confident in the other AI. When a designer accepts or rejects both AI suggestions and receives positive feedback, as expected, the confidence in AI #1 and in AI #2 increases and decreases together, respectively. In contrast, a designer's self-confidence slightly increases with any positive feedback (with a small exception explained in the following paragraph). This result coincides with the results from [25] and suggests that a human designer judges their own task ability primarily based on the success of their final action (i.e., the feedback) during AI-assisted decision-making.

The exceptions of the results from the previous paragraph provide significant insights. The exception scenarios are when positive feedback is given to one of the AIs in the Hi–Lo condition, in which the impact of the feedback on a human designer's confidence is trivial. As shown in the results described in the previous paragraph, unlike negative feedback which is shown to impact all confidence negatively, positive feedback is attributed to the specific subject of feedback. Therefore, the trivial impact of positive feedback in the Hi–Lo condition may be because of some implicit confusion about the difference between the two AIs' performance and therefore which of them to attribute more confidence to.

The different effects of positive and negative feedback on a human designer's confidence in their AI teammates and in themselves inform the choice and design of feedback to improve the outcome of AI-assisted decision-making. For example, negative feedback must be provided with prudence as it is detrimental to a human designer's confidence and trust in their AI teammates as well as in themselves. According to the results of this work, this prudence is especially important with two-AI teammates where it can be harder to increase a human designer's confidence in AI and self-confidence than with one AI. Additionally in design practice, it would help for the team or its manager to be aware that positive feedback on an AI teammate causes a human designer to be less confident in the other AI teammate. Based on these insights, appropriate feedback can be provided to a human designer to guide their confidence levels. Finally, the findings in this section can be used to identify potential reasons when very low or high confidence levels are detected.

4.3 How Are These Confidences Each Related to a Human Designer's Probability of Accepting Artificial Intelligence #1's Suggestions and Probability of Accepting Artificial Intelligence #2's Suggestions? Having discussed the impact of the performance combination of the two-AI teammates and feedback on a human designer's confidence in each AI and self-confidence, this section discusses the impact of this confidence on their selection between one of the AI's or their own design action. Despite the intuitive hypothesis that all confidence will affect this selection, only the confidence in AI #1 and self-confidence have a significant impact. Recalling [25] one-human/one-AI teams which found that both confidence in the AI and self-confidence impact a human designer's acceptance of AI suggestion, the weak to no impact of confidence in AI #2 in this study suggests that a human designer does not have the cognitive capacity (i.e., cognitive overload [45]) to consider their confidence for another AI during this selection. There are two possible reasons for considering confidence in AI #1 over AI #2 rather than the other way around: the order of presentation of the two-AI suggestions (i.e., AI #1 on the left and AI #2 on the right) and/or the lack of a Lo–Hi condition in the study (i.e., the overall experimental dataset is biased towards AI #1 being higher performing than AI #2). The reason cannot be concluded without conducting a future study with a Lo–Hi condition.

How does a designer's confidence in AI #1 and self-confidence influence the probabilities of accepting either one of the AI suggestions differ between the experimental conditions. In the Hi–Hi condition, a designer's confidence in AI #1 rightly is positively correlated with their probability of accepting AI #1's suggestion and is negatively correlated with their probability of accepting AI #2's suggestion. Oddly, however, these correlations are in opposite directions in the Lo–Lo condition. These findings suggest that because a designer is more likely to have received more positive feedback in the Hi–Hi condition, they may be prone to rely on their own judgment of the AIs (i.e., confidence in the AIs) during the final action selection. However, receiving negative feedback in the Lo–Lo condition, a designer may choose to go against their own judgment of the AIs. Unlike the other two conditions, in the Hi–Lo condition, a designer's confidence in AI #1 is positively correlated with both probabilities of accepting AI #1's or AI #2's suggestion. This may insinuate again that it is difficult for designers to distinguish between the two AIs' performances in this condition, therefore applying their confidence in AI #1 to the decisions to accept AI #1 suggestions and to accept AI #2 suggestions. Strikingly, in all three conditions, self-confidence impacts the action

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selection in the opposite direction as confidence in AI #1 does. For example, in the Hi–Hi condition, confidence in AI #1 is positively correlated to the probability of accepting AI #1's suggestions, and self-confidence is negatively correlated to this probability.

Understanding the relationships between a human designer's confidence in the two AIs and self-confidence and their final action selection presents opportunities to effectively manage a human designer to make good decisions, resulting in successful team performance. For example, given the finding that confidence in AI #1 and self-confidence influence a human designer's final decision, targeted feedback can be provided to a human designer based on the results from RQ2 to guide their confidence in AI #1 and self-confidence, resulting in desirable acceptance or rejection of AI suggestions.

4.4 Is a Human Designer More Likely to Accept a Design Suggestion That Two Artificial Intelligence s Agree on Than One Provided by a Single AI? Finally, this work investigates whether a human designer is more likely to conform to an AI suggestion in a one-human/two-AI team decision-making scenario than in a one-human/one-AI team scenario [25]. The results in Table 4 show no significant difference in the probability of accepting an AI suggestion between the two scenarios, rejecting the earlier hypothesis that an additional AI will increase information overload and as a result the likelihood of a human designer conforming to an AI suggestion [39]. Furthermore, it is found that a human designer is not more likely to conform to an AI suggestion even when the two-AI teammates agree on a suggestion. In other words, a human designer is not likely to conform to the majority opinion when making decisions in a one-human/two-AI design team. One possible explanation for this result is the small size of the majority group (i.e., two). Asch and Hogg and Vaughan found that the maximum conformity effects are found when the size of the majority is three to five [46,47]. Therefore, two agreeing AI teammates may not be enough in this scenario to cause a significant increase in a human designer's conforming behavior.

The finding that an additional AI does not increase the possibility of accepting an AI suggestion suggests that it is futile in practice to implement another AI into a human–AI design team merely to gain more trust from a human designer. Combined with the insights from the rest of this work, an additional AI (i.e., two-AI teammates) is likely to lead to decreases in a human designer's confidence in AI and self-confidence. Therefore, unless there is a clear advantage of having another AI teammate, implementing multiple AIs in a human–AI design team must be done with extra caution.

4.5 Limitations and Areas of Future Work. There are some limitations in this work that bring about exciting opportunities for future work. First, this work cannot conclude why a human designer considers their confidence in AI #1 over their confidence in AI #2 during their final action selection. To gain insight into this finding, an additional study comparing the Hi–Lo condition to a Lo–Hi condition where AI #2's suggestion is provided on the left of AI #1's suggestion could be conducted. Another potential shortcoming of this work is that the confidence data are self-reported. Although it is a commonly used method of measuring confidence, the accuracy of this method has been debated [48–50]. Kvam et al. also found that explicitly repeated reporting influences people's subsequent judgments [51]. Therefore, reproducing this study with other types of confidence measurements can help confirm the results of this work. Additionally, the experiment in this work has been designed with AIs that solve and provide suggestions for the same problem. Future work on complementary AIs that provide suggestions regarding different aspects of the problem would be interesting. Lastly, it is important to realize that the insights from this work may not be generalizable to all design scenarios and tasks. Therefore, exploring how the results from this work apply to areas such as conceptual design or need finding would be a significant line of future work.

5 Conclusion

This work utilizes a human subjects experiment and a quantitative model to examine a human designer's confidence in their AI teammates, self-confidence, and acceptance of AI input during an AI-assisted decision-making, engineering design task with two-AI teammates. It is critical to study such AI-assisted decision-making scenarios with more than one AI because multiple AIs are likely to be increasingly deployed to design teams to solve complex, multifaceted design problems. This work specifically explores and discusses how a human designers' confidence changes with different AI teammates' performance combinations and various types of trial-by-trial experiences and feedback, as well as how these confidences relate to their decision to accept or reject the AI suggestions. This work conducts its discussion of the results in comparison to a prior study on AI-assisted decision-making scenarios with one-AI teammate and notes the impact of an additional AI. Overall, this work offers a better understanding of human designers' decision-making process given suggestions from two-AI teammates. This understanding presents opportunities to strategically design human–AI teams (e.g., team size, the performance of the members) and guide a human designer's decision to accept or reject AI suggestions. These opportunities can facilitate strategies for AI-assisted decision-making in design.

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Conflict of Interest

There are no conflicts of interest.

Data Availability Statement

The datasets generated and supporting the findings of this article are obtainable from the corresponding author upon reasonable request.

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